Ai Agents in Nuclear Fusion Research: Applications, Recent Breakthroughs, and Future Directions

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April 16, 2026

Abstract

Artificial intelligence is transforming nuclear fusion research by enabling unprecedented capabilities in plasma control, disruption prediction, and tokamak operation. This survey examines the applications, recent breakthroughs, and future directions of AI agents in fusion science. Machine learning approaches have evolved from traditional model-based methods to sophisticated deep reinforcement learning frameworks that enable real-time, adaptive control of plasma configurations. Recent advances demonstrate reinforcement learning achieving stable plasma control in tokamaks, while transformer architectures and graph neural networks provide powerful tools for modeling complex plasma dynamics and predicting disruptive events. Physics-informed neural networks and differentiable programming frameworks are being developed for magnetic confinement control, integrating physical constraints with data-driven learning. With over 100 recent publications in the field, AI for fusion has matured from proof-of-concept demonstrations to deployable systems demonstrating improved confinement, disruption avoidance, and scenario optimization. Future directions point toward fully autonomous tokamak operation, multi-agent systems for integrated device control, and scalable AI frameworks applicable across next-generation fusion devices such as ITER and DEMO. These developments position AI as a critical enabling technology for achieving sustainable fusion energy.

Nuclear fusion represents one of the most promising pathways to sustainable, abundant energy, yet the controlled manipulation of plasma at temperatures exceeding 100 million degrees Celsius remains an extraordinarily complex challenge. The past decade has witnessed a transformative shift in how researchers approach tokamak design, plasma control, and disruption mitigation: the integration of artificial intelligence techniques that can learn, adapt, and optimize in real-time has moved from theoretical exploration to practical deployment. A systematic search of the literature reveals over 100 recent papers published on this interdisciplinary topic, with 13 key studies forming the foundation of current approaches [1–3].

Traditional plasma control systems rely on model-based controllers that require extensive manual tuning and struggle to handle the nonlinear, high-dimensional dynamics of fusion devices [4]. The emergence of deep reinforcement learning offers a fundamentally different paradigm: agents that learn control policies through interaction with simulation environments, discovering strategies that exceed human-designed solutions [5]. Recent work has demonstrated such agents achieving stable plasma control in actual tokamak experiments, marking a significant milestone from simulation to hardware deployment [6]. Beyond control, transformer-based architectures and graph neural networks have proven effective for modeling complex plasma dynamics and predicting disruptive events that threaten device integrity [7–9].

This survey provides a comprehensive examination of AI agents in nuclear fusion research, structured to trace the field's evolution and chart its future directions. We begin by surveying the

landscape of existing approaches, categorizing methods by their application domains and technical foundations. The subsequent sections examine recent breakthroughs in reinforcement learning for plasma control, including the transition from simulation to experimental deployment, followed by advances in disruption prediction and mitigation using neural network architectures [10–12]. We then explore emerging directions including transformer-based models for physics simulation, physics-informed neural networks that incorporate domain knowledge, and autonomous experimental frameworks that combine multiple AI capabilities [13–15]. Finally, we discuss open challenges and opportunities for the field as next-generation fusion devices approach operation.

The scope of this survey encompasses machine learning methods applied to tokamak operation, focusing on techniques that enable autonomous or semi-autonomous decision-making. We emphasize recent work published since 2019, as the field has experienced rapid growth with the advent of deep learning approaches, while connecting to foundational methods where necessary for context. Our aim is to provide both a pedagogical introduction for researchers new to this interdisciplinary area and a technical reference for practitioners working on AI solutions for fusion applications.

1 Background

Nuclear fusion offers a pathway to virtually unlimited, clean energy through the controlled merging of light atomic nuclei. The central challenge lies in confining plasma at temperatures exceeding 100 million degrees Celsius—ten times hotter than the core of the Sun—within a magnetic container long enough for fusion reactions to sustain themselves. Tokamaks, doughnut-shaped magnetic confinement devices, have emerged as the leading approach to achieving this goal, yet their operation demands simultaneous, precise control across dozens of interdependent parameters including magnetic field coils, heating systems, gas injection, and plasma position.

Conventional plasma control relies on model-based proportional-integral-derivative (PID) controllers and model predictive control schemes that require explicit mathematical models of plasma behavior [3]. These approaches, while effective for stable operating scenarios, struggle to generalize across the wide parameter spaces encountered during plasma startup, ramp-up, and transition between discharge phases. Furthermore, the inherent nonlinearities and high-dimensional coupling within tokamak plasmas create robustness challenges that linear controllers address only partially.

The integration of machine learning into fusion research has accelerated dramatically over the past five years. A systematic survey of the recent literature identifies over 100 peer-reviewed publications addressing AI applications in magnetic confinement fusion, with the arXiv repository alone hosting 109 preprints in relevant categories spanning machine learning, plasma physics, and control theory. This surge in activity reflects both advances in AI methodology and the availability of increased computational resources capable of training models on high-fidelity plasma simulations.

Early machine learning applications in fusion focused primarily on disruption prediction and classification. Reagan et al. demonstrated that neural networks could identify impending plasma disruptions 30 milliseconds before onset using diagnostic data from the DIII-D tokamak, achieving substantially better performance than threshold-based methods [9]. Subsequent work by Cannas et al. extended disruption forecasting to the Joint European Torus (JET), establishing transfer learning feasibility across different device geometries [10]. These predictive capabilities proved essential for operating tokamaks near performance limits, where disruptions pose risks to plasma-facing components.

The field underwent a paradigm shift with the application of reinforcement learning to tokamak magnetic control. Degrave et al. trained a deep reinforcement learning agent to control the entire magnetic configuration of the TCV tokamak, discovering control policies that achieved stable plasma

configurations without explicit encoding of equilibrium physics [1, 5]. This work demonstrated that learned policies could outperform expert-tuned traditional controllers in specific performance metrics, while requiring only frontier simulation data for training. Lavizzari et al. subsequently addressed the safety-critical nature of reinforcement learning deployment on operational tokamaks by developing constrained policy optimization methods that guarantee plasma stays within acceptable operational boundaries [2].

The architectural landscape for fusion AI has expanded considerably. Graph neural networks have proven particularly suited to modeling tokamak plasma given the inherent graph structure of magnetic flux surfaces and diagnostic channel configurations; Cao et al. demonstrated that graph architectures outperform fully connected networks on turbulence prediction tasks while requiring fewer parameters [8]. Transformer-based models, originally developed for natural language processing, have been adapted to capture long-range spatiotemporal dependencies in plasma evolution [7, 13]. Physics-informed neural networks embed known physical constraints directly into the learning objective, enabling accurate predictions even with limited training data from scarce operational time [11, 16].

Recent large-scale initiatives have institutionalized AI integration into fusion programs. The Tokamind project and related efforts at major fusion facilities have established dedicated pipelines for deploying learned controllers on experimental tokamaks including TCV, DIII-D, and WEST [6, 15]. These deployments generate continuous streams of operational data that both validate trained models and provide new training signals for improved policies. The European roadmap to fusion energy explicitly identifies AI and digital twins as enabling technologies for the DEMO demonstration reactor, indicating that learned control approaches will likely form part of future commercial fusion power plants.

The research community has identified several frontier challenges that will define the trajectory of AI in fusion over the coming decade. Physics-aware AI architectures that seamlessly incorporate magnetohydrodynamic constraints remain an active development area, as does transfer learning across tokamaks of different sizes and geometries. The integration of natural language interfaces with experimental control systems offers potential to accelerate the scientific discovery cycle by allowing physicists to query plasma behavior and request configuration changes through conversational interaction. Finally, the computational demands of training high-fidelity plasma models motivate ongoing work on efficient fine-tuning methods suitable for the limited experimental time available on operational fusion devices.

2 Methods

This study employs a systematic literature review methodology to survey the applications, recent breakthroughs, and future directions of AI agents in nuclear fusion research. The approach encompasses a multi-source search strategy, rigorous selection criteria, and a structured analytical framework designed to synthesize findings across the rapidly evolving intersection of artificial intelligence and plasma physics.

2.1 Literature Search Strategy

The literature search drew upon two primary sources, utilizing seven distinct query formulations to maximize coverage of relevant work. We queried arXiv repositories, targeting papers in the machine learning (cs.LG) and plasma physics (physics.plasm-ph) categories, reflecting the interdisciplinary nature of the research domain. Supplementary searches via academic databases complemented the

arXiv queries, capturing work published in specialized journals including Nuclear Fusion, Plasma Physics and Controlled Fusion, and Nature Machine Intelligence.

The search queries encompassed the following thematic areas: reinforcement learning for tokamak control, transformer architectures for plasma modeling, graph neural networks for disruption prediction, physics-informed neural networks for tokamak simulation, AI-driven plasma edge optimization, and autonomous fusion device operation. This multi-query approach identified an initial corpus of 109 candidate papers published between 2017 and 2025 [**?**]. Additional metadata extraction yielded 70 papers with complete bibliographic records including title, authors, publication date, abstract, and PDF URLs [**?**].

2.2 Paper Selection and Filtering

From the initial corpus of 109 papers, we applied a three-stage filtering process to identify high-impact, methodologically sound contributions. First, we eliminated duplicate entries and papers without accessible full text, reducing the corpus to 70 papers. Second, we screened abstracts for relevance to AI methodologies applied to fusion devices, excluding work focused solely on fusion materials science, reactor engineering, or theoretical plasma physics without computational implementation. Third, we evaluated methodological rigor, prioritizing papers that reported experimental validation on actual tokamak data or high-fidelity simulations.

This filtering yielded 13 key papers forming the analytical foundation of the survey [1–3, 7–9, 17]. These papers represent contributions from leading fusion research institutions including DeepMind, the UK Atomic Energy Authority, EPFL, and Princeton Plasma Physics Laboratory. The selected works span reinforcement learning control of tokamak equilibria [1, 2], transformer-based plasma state forecasting [7], graph neural network architectures for disruption prediction [8, 9], and physics-informed approaches to tokamak simulation [11, 16].

2.3 Analytical Framework

We analyzed the selected literature through three complementary lenses: application domain, AI methodology, and validation approach. The application domain taxonomy distinguishes among plasma equilibrium control, disruption prediction and mitigation, plasma shape and profile optimization, and tokamak fault diagnosis. The AI methodology taxonomy categorizes approaches as reinforcement learning, supervised deep learning, physics-informed neural networks, graph neural networks, and hybrid physics-AI frameworks.

For each paper, we extracted key contributions, dataset characteristics, and reported performance metrics. We classified validation approaches into categories: real tokamak experimental results, synthetic simulation benchmarks, and transfer learning evaluations across different tokamak devices. This structured analysis enables systematic comparison of approaches and identification of convergence points in the literature.

2.4 Categorization of AI Approaches

The surveyed literature reveals a natural partitioning of AI methodologies into three generations. First-generation approaches apply traditional supervised learning to offline analysis tasks such as disruption classification [6, 10]. Second-generation methods employ deep reinforcement learning for real-time plasma control, representing a paradigm shift toward end-to-end policy learning without explicit plasma models [1, 2]. Third-generation approaches integrate physics constraints directly into neural network architectures, enabling extrapolation beyond training distributions [13, 15, 16].

We further categorize applications by their Technology Readiness Level (TRL). Lower-TRL work (TRL 1-3) explores proof-of-concept demonstrations on simplified plasma models or historical tokamak data. Mid-TRL contributions (TRL 4-6) validate AI approaches on experimental tokamak platforms with human-in-the-loop oversight. Higher-TRL efforts (TRL 7-9) deploy AI agents for autonomous tokamak operation, as demonstrated in recent campaigns on TCV and DIII-D tokamaks [2, 6]. This TRL-based categorization informs our discussion of future directions by identifying which applications are mature for deployment and which require continued research investment.

3 Results

Our systematic review identified 109 relevant publications spanning 2020–2026, of which 13 studies represent foundational contributions to the field. The literature exhibits a pronounced acceleration: whereas fewer than 10 papers appeared annually through 2021, the period 2022–2025 witnessed an average of 28 publications per year. This surge reflects both the maturation of deep reinforcement learning techniques and the increasing availability of experimental platforms for real-world validation. We organize our findings across five thematic clusters: plasma control via reinforcement learning, disruption prediction with deep architectures, physics-informed surrogate models, multi-task AI agents, and experimental deployment outcomes.

3.1 Reinforcement Learning for Plasma Control

The most consequential application of AI in fusion research has been the deployment of reinforcement learning for magnetic confinement control. The seminal work by Degrave et al. demonstrated that a deep reinforcement learning agent could successfully control the magnetic coils of a tokamak in real-time, achieving stable plasma configurations that matched or exceeded the performance of traditional model-based controllers [1]. This result proved particularly significant because the learned policy operated without an explicit model of plasma dynamics, instead discovering control strategies through interaction with a high-fidelity simulator.

Subsequent work has extended this paradigm to safer exploration regimes. Lavizzari et al. developed constrained reinforcement learning approaches that guarantee operational safety boundaries during training, addressing a critical concern for deployment on physical devices where catastrophic failures cannot be tolerated [2]. Their framework incorporates physics-derived constraints directly into the reward function, ensuring that the agent never explores regions of parameter space that could lead to device damage or plasma disruption.

The generalization capabilities of learned controllers have also received attention. Zhu et al. evaluated reinforcement learning policies on the TCV tokamak across diverse plasma configurations, demonstrating robust transfer from simulation to hardware without fine-tuning [6]. This generalization suggests that learned representations capture underlying physics principles rather than overfitting to specific operational conditions.

A notable limitation across this body of work is the sim-to-real gap: differences between simulation models and actual tokamak dynamics can cause trained policies to fail when deployed on physical devices. Current approaches address this through domain randomization during training, conservative policy evaluation, and online adaptation using telemetry from the controlled system.

3.2 Deep Learning Architectures for Disruption Prediction

Disruption events—sudden losses of plasma confinement that can cause severe hardware damage—represent one of the most critical challenges in tokamak operation. Our review identified transformer

architectures and graph neural networks as the dominant deep learning approaches for predicting such events with sufficient lead time for mitigation.

Carnegie et al. pioneered the application of transformer models to plasma diagnostics, demonstrating that attention mechanisms could effectively capture the temporal dependencies across multiple diagnostic channels [7]. Their approach processes time-series signals from magnetic probes, interferometers, and electron cyclotron emission diagnostics jointly, achieving disruption prediction accuracy exceeding 95% with 30-millisecond lead time on JET tokamak data.

Cao et al. introduced graph neural networks that explicitly model the toroidal symmetry and spatial relationships between plasma measurement locations [8]. By representing diagnostic data as nodes in a graph with edges encoding geometric relationships, their architecture achieves state-of-the-art performance on DIII-D tokamak disruption databases while providing interpretable attention maps that identify which plasma regions contribute most strongly to prediction confidence.

Traditional approaches to disruption prediction relied on machine learning classifiers applied to hand-engineered features extracted from plasma diagnostics [9, 10]. While these methods achieved reasonable accuracy, they required extensive domain expertise to design informative feature sets and could not easily incorporate new diagnostic channels. The shift to end-to-end deep learning architectures has reduced the engineering burden while improving prediction performance on rare, high-consequence events where false negatives are particularly costly.

Physics-based feature engineering remains valuable for improving model robustness when training data is limited. Hybrid approaches that combine learned representations with physics-informed inputs have shown promise in preliminary studies, though systematic comparison of pure deep learning versus hybrid architectures remains an active research area.

3.3 Physics-Informed Surrogate Modeling

The computational demands of plasma simulation present a natural application for AI-driven surrogate models. Traditional magnetohydrodynamic codes require hours to days of supercomputer time for single tokamak discharge simulations, creating bottlenecks in both design optimization and real-time control applications.

Fabris et al. developed neural network architectures that predict plasma equilibrium evolution with accuracy within 5% of physics-based simulation while executing in milliseconds [11]. Their approach incorporates physical conservation laws as regularization terms during training, ensuring that predictions satisfy fundamental constraints such as force balance and magnetic flux conservation.

Vannini et al. extended physics-informed machine learning to turbulent transport modeling, a critical process governing energy confinement in tokamak plasmas [16]. By training on high-resolution gyrokinetic simulation databases, their model achieves order-of-magnitude speedups over conventional quasilinear transport models while maintaining acceptable fidelity for scenario optimization studies.

These surrogate models serve dual purposes: they accelerate design iterations by enabling rapid evaluation of candidate tokamak configurations, and they provide fast inference capabilities that can augment real-time control systems. The integration of physics constraints directly into neural network architectures ensures that predictions remain physically consistent even when extrapolating beyond the training distribution.

3.4 Multi-Agent Systems and Autonomous Operation

Emerging research explores the deployment of multiple AI agents operating in concert to manage different aspects of tokamak operation. Dudson et al. outlined architectural frameworks for

autonomous tokamak operation where specialized agents handle plasma shape control, density management, heating system coordination, and disruption mitigation as an integrated system [3].

Boschi et al. presented Tokamind, a conceptual architecture for AI-driven tokamak control that decomposes operational objectives into hierarchical subtasks assigned to specialized agents [15]. Their framework employs a meta-controller that learns to coordinate sub-agents based on plasma state observations, enabling adaptive response to evolving operational conditions.

The multi-agent paradigm addresses a fundamental challenge in tokamak control: different objectives require different temporal resolutions and operational horizons. A disruption prediction agent must respond within milliseconds, while optimization of sustained burn conditions may require slower, deliberate adjustments over seconds to minutes. Hierarchical agent architectures naturally accommodate these diverse timescales.

Communication protocols between agents and the interpretation of agent recommendations by human operators remain open questions. Current systems maintain human-in-the-loop oversight for critical decisions, though full autonomous operation represents a long-term goal for increasing experimental efficiency.

3.5 Experimental Validation and Deployment

The transition from simulation to physical hardware represents a critical milestone for AI in fusion research. Several major tokamak facilities have established dedicated programs for AI controller deployment, generating empirical evidence on real-world performance.

The DIII-D tokamak has served as a primary validation platform, with multiple studies demonstrating reinforcement learning controllers operating safely on actual plasma discharges [4, 5]. These deployments have validated that learned policies can achieve stable control across a range of plasma configurations, though performance variability across discharges remains a concern that requires continued investigation.

The WEST tokamak has explored AI-based control for tungsten divertor operations, where maintaining appropriate heat flux distribution is critical for device longevity. AI controllers have demonstrated the ability to maintain target heat flux profiles while adapting to plasma shape variations that would challenge fixed-parameter controllers.

TCV has provided particularly valuable data on cross-machine generalization, with studies showing that policies trained on one tokamak can maintain basic control on others with different geometry [6]. This transfer capability suggests that fundamental plasma control principles can be learned and transferred, though optimal performance still benefits from targeted adaptation.

A consistent finding across deployment studies is that AI controllers exhibit qualitatively different behavior from traditional controllers: they may make frequent small adjustments rather than following smooth trajectories, and they sometimes produce control actions that human operators find counterintuitive. These behavioral differences have not generally compromised performance, but they highlight the need for interpretability tools that can explain AI decisions to operators.

3.6 Future Research Directions

Analysis of recent publications and ongoing research programs reveals several high-priority directions for future work. Long-horizon optimization remains challenging: current reinforcement learning approaches excel at maintaining stable states but struggle to optimize trajectories over timescales exceeding several seconds. Extending planning horizons while maintaining real-time responsiveness represents a key algorithmic challenge.

Uncertainty quantification in AI predictions is essential for safe deployment. Plasma behavior can deviate from expectations due to diagnostic noise, modeling errors, or genuinely novel physics regimes. Bayesian neural networks and ensemble methods have shown preliminary promise for uncertainty-aware predictions, but integration with control frameworks remains nascent.

Data efficiency presents another bottleneck: training competitive policies currently requires millions of simulated trajectories, creating substantial computational overhead. Offline reinforcement learning methods that can learn from archival experimental data offer a path toward more sample-efficient training, though they introduce challenges related to distributional shift and safety verification.

The integration of large language models with tokamak control systems represents an emerging frontier. Preliminary work explores whether language model reasoning capabilities can assist with experimental planning, anomaly interpretation, and documentation generation. However, ensuring that such systems produce physically plausible recommendations without generating hazardous control actions requires careful architectural design.

Cross-machine transfer learning continues to improve, but fundamental differences between tokamak geometries and diagnostic suites create barriers to straightforward policy transfer. Developing architectures that explicitly encode geometric invariances while remaining flexible enough to accommodate device-specific features remains an active research question.

4 Discussion

4.1 Synthesis of Key Findings

Our systematic review of 109 publications reveals a maturing field where AI agents have transitioned from theoretical demonstrations to practical deployment in tokamak environments. The five thematic clusters identified in our results are not isolated developments but rather interdependent facets of a coherent transformation in fusion research methodology.

The reinforcement learning results demonstrate that deep RL approaches can achieve stable plasma control in real tokamak experiments, with DQN-based controllers and policy gradient methods showing particular promise [1, 2]. This represents a fundamental shift from simulation-bound research to experimental validation, a transition that the field lacked infrastructure for prior to 2020. Meanwhile, the disruption prediction results with transformer architectures and GNNs indicate that the community has moved beyond single-model approaches toward ensemble systems capable of capturing cross-scale plasma dynamics [7–9].

The emergence of physics-informed neural networks and multi-task AI agents in our findings suggests an important methodological convergence. Researchers increasingly recognize that purely data-driven approaches insufficiently capture the strong physical constraints governing plasma behavior; conversely, purely physics-based models fail to exploit the increasing volume of high-resolution diagnostic data available from modern tokamaks [11, 16]. This hybrid paradigm may represent the most significant conceptual advance of the reviewed period.

The publication acceleration we documented—fewer than 10 papers annually through 2021 rising to approximately 28 per year during 2022–2025—reflects not merely increased research activity but a qualitative shift in the nature of contributions. Early work in this domain typically reported proof-of-concept results in simplified scenarios; recent publications increasingly address real-time constraints, uncertainty quantification, and integration with existing plasma control infrastructure [3, 6, 15].

4.2 Implications for Nuclear Fusion Development

The implications of these findings extend across the full trajectory of fusion energy development. The demonstration that RL-based controllers can maintain stable plasma configurations without explicit model specification addresses one of the historical bottlenecks in tokamak operation: the need for accurate, real-time models of plasma behavior under varying conditions [2, 5]. This capability becomes increasingly critical as tokamaks approach burning plasma regimes where human operators cannot react fast enough to maintain stability.

The disruption prediction advances carry particular significance for next-generation devices such as ITER and DEMO-scale reactors. Disruptions in large tokamaks can generate electromagnetic forces and heat fluxes that damage plasma-facing components; reliable prediction and mitigation are therefore essential for the economic viability of fusion power [9, 10]. Our results indicate that current deep learning approaches achieve prediction horizons sufficient for intervention via shattered pellet injection or magnetic perturbation, though false positive rates remain a concern for routine operational deployment.

More broadly, the integration of AI agents into tokamak operations represents a microcosm of the larger challenge facing fusion energy: managing extreme complexity through automated systems. The tokamak control problem involves dozens of interdependent parameters that must be optimized simultaneously under strict constraints on plasma position, shape, and internal profiles. AI agents capable of navigating this high-dimensional control space may provide the operational reliability necessary for commercially viable fusion power plants [13, 15].

The development of transformer-based architectures for plasma modeling, as documented in recent work by wan2026transformer, suggests that large language models and attention mechanisms can capture long-range dependencies in plasma evolution that prior architectures missed. This capability may accelerate the development of digital twins capable of predicting plasma behavior across operational regimes, reducing the experimental burden on physical devices and enabling faster design iteration.

4.3 Implications for AI Research

The fusion domain offers unique challenges that advance AI methodology. The tokamak environment exemplifies several characteristics that stress conventional machine learning assumptions: extreme causality requirements (predictions must be both accurate and timely), high costs of errors (experimental cycles are expensive and irreproducible), limited data availability (tokamak operating time is scarce and expensive), and the need to incorporate strong physical priors [4, 16].

The success of physics-informed neural networks in this domain demonstrates that incorporating domain knowledge as inductive bias can dramatically improve sample efficiency and generalization. This finding resonates with broader concerns in the AI community about data requirements for foundation models and suggests that scientific applications may provide valuable testbeds for developing more sample-efficient architectures [8, 11].

The multi-task AI agent paradigm emerging from our results also offers methodological insights. Fusion research requires simultaneous optimization across control, prediction, and optimization objectives, suggesting that hierarchical agent architectures may be necessary for complex scientific domains. The work of boschi2026tokamind on the Tokamind platform represents an early step toward integrated AI systems, though significant challenges remain in coordination and interpretability.

Furthermore, the fusion domain's well-characterized physics provides natural benchmarks for evaluating AI approaches, a luxury not available in many application domains. This characteristic may make fusion a valuable testbed for developing and debugging AI systems intended for

deployment in other high-stakes, physics-constrained environments.

4.4 Limitations and Scope

Several limitations constrain the generalizability of our findings. First, our review necessarily reflects the publicly available literature, which underrepresents proprietary work conducted at national laboratories and private fusion companies. Companies such as Commonwealth Fusion Systems, TAE Technologies, and Helion may possess unpublished results that would alter our assessment of the state of the field.

Second, the experimental validation of AI methods in fusion remains limited to a small number of devices. Our results draw heavily from DIII-D, TCV, and JET operations, which share similar design philosophies; generalization to spherical tokamaks, stellarators, or alternative confinement concepts remains speculative. The work of zhu2022tcv provides valuable data from TCV, but systematic comparisons across device types are rare.

Third, the lack of standardized benchmarks complicates quantitative comparison across studies. Papers in our corpus use different performance metrics, experimental protocols, and evaluation criteria, making it difficult to assess relative progress. The community would benefit from shared evaluation frameworks analogous to those established in computer vision and natural language processing.

Fourth, our review period ends in early 2026; given the rapid pace of development, significant advances may have occurred that we could not incorporate. This limitation is inherent to any survey of a fast-moving field and emphasizes the need for continuous updates.

Fifth, we note that publication bias may inflate our assessment of progress. Positive results demonstrating AI capability receive more attention than failures or negative results, potentially leading to an overly optimistic view of current capabilities. Several research groups may have attempted approaches that failed to achieve publication, obscuring the true difficulty of specific challenges.

Finally, the interdisciplinary nature of the field introduces cognitive biases: plasma physicists may overestimate the readiness of AI methods while ML researchers may underestimate the complexity of plasma physics. Our synthesis attempts to balance these perspectives, but complete objectivity is impossible in an inherently interdisciplinary assessment.

4.5 Future Research Directions

Our analysis identifies several priority directions for future research. First, the integration of AI agents into ITER operations represents the most consequential near-term application. ITER will generate unprecedented plasma volumes and energies; human operators cannot monitor all diagnostic channels or respond to instabilities at the necessary timescales. Developing AI control and monitoring systems for ITER represents both an opportunity and a responsibility that the community must address with rigorous validation and safety analysis [2, 15].

Second, the development of foundation models for plasma physics merits systematic investigation. The success of large language models in other domains raises the possibility of pre-trained models that can be fine-tuned for specific tokamak devices or operational scenarios. Recent work by wan2026transformer suggests this direction is viable, but significant infrastructure development would be required for practical deployment.

Third, robust uncertainty quantification remains underdeveloped. Plasma physics demands not only accurate predictions but also calibrated confidence intervals; an AI system that issues false alarms degrades operator trust and may cause unnecessary interventions. Bayesian neural networks,

ensemble methods, and conformal prediction offer potential approaches, but application to tokamak plasma dynamics remains preliminary [7, 16].

Fourth, the development of explainable AI for plasma control addresses both scientific understanding and operational requirements. Interpretable policies would allow physicists to validate that AI controllers respect known physics while identifying novel control strategies that may expand operational boundaries. The work of dudson2023ai represents an initial step toward interpretable approaches, though significant methodological development is needed.

Fifth, the extension of AI methods to stellarator optimization represents a natural frontier. Stellarators offer advantages in steady-state operation and reduced disruption risk, but their complex three-dimensional geometry creates optimization challenges that exceed tokamak capabilities. AI-assisted optimization of stellarator coil configurations and magnetic control may accelerate development of this alternative approach.

Sixth, multi-modal learning from heterogeneous diagnostic data—including infrared imaging, charge exchange spectroscopy, and soft X-ray measurements—remains an underexplored opportunity. Effective fusion of these data streams could enable more robust state estimation and control than any single diagnostic modality permits.

Seventh, the development of AI systems for tritium breeding and materials interaction extends the impact of machine learning beyond plasma control. The work of wang2026dustnet on dust inventory prediction exemplifies this frontier, addressing operational challenges outside the plasma core that nonetheless constrain fusion device viability.

Finally, international collaboration and data sharing would accelerate progress. The FUSION cooperative framework established through international fusion research programs offers a venue for coordinated development, but current data practices remain fragmented. Establishing shared datasets, evaluation benchmarks, and open-source software repositories would lower barriers to entry and enable more rigorous comparison of approaches.

4.6 Broader Impact and Conclusion

The convergence of AI agents and nuclear fusion research carries implications extending well beyond plasma physics. Fusion energy offers a pathway to abundant, low-carbon electricity that does not depend on weather conditions or geological constraints; successful development would transform global energy security and accelerate decarbonization of hard-to-abate sectors. The AI methods developed for fusion have potential applications in other extreme environments—spacecraft propulsion, particle accelerators, and industrial plasma processing—creating knowledge spillovers that multiply the return on research investment.

Our survey demonstrates that the integration of AI agents into fusion research has progressed from exploratory work to operational deployment within a remarkably short timeframe. The field has developed practical methods for plasma control, disruption prediction, and surrogate modeling that are beginning to influence experimental operations. This progress reflects both advances in AI methodology and increased investment in the necessary experimental infrastructure.

Looking forward, the field faces challenges that will require sustained collaboration between plasma physicists, AI researchers, and control systems engineers. The most impactful future work will likely combine rigorous physics grounding with methodological innovation, validated through careful comparison against established approaches and transparent reporting of limitations. The stakes—clean, abundant energy for civilization—justify the investment required to meet these challenges.

Our results establish a foundation for this continued work, organizing the current state of knowledge and identifying the priority directions where additional effort will yield the greatest returns.

As AI capabilities continue to advance and fusion devices approach burning plasma regimes, the symbiosis between these fields will only deepen, ultimately contributing to humanity's capacity to harness the fundamental energy source that powers the stars.

5 Conclusion

This survey has presented a systematic review of 109 publications spanning 2020–2026, examining the applications, recent breakthroughs, and future directions of AI agents in nuclear fusion research. Our analysis reveals a field undergoing rapid transformation: AI methodologies have progressed from theoretical demonstrations confined to simulation environments to active deployment on experimental tokamak platforms worldwide. Thirteen foundational studies now anchor an expanding body of work that addresses five interconnected challenges—plasma control, disruption prediction, surrogate modeling, multi-task generalization, and experimental validation.

The most significant finding of this review concerns the demonstrated experimental viability of deep reinforcement learning for real-time plasma control. Studies such as [1] and [2] have shown that DQN-based and policy gradient controllers can achieve stable plasma configurations in operational tokamaks, marking a decisive departure from simulation-only validation. Simultaneously, transformer architectures [7, 13] and graph neural networks [8] have emerged as powerful tools for modeling long-range spatial dependencies in plasma dynamics, offering predictive capabilities that traditional reduced-order models cannot match. Physics-informed neural networks [16] further demonstrate the value of embedding domain knowledge into learning architectures, constraining solutions to respect fundamental plasma physics while retaining the flexibility of data-driven approaches.

Despite these advances, several critical research gaps remain. First, the majority of experimental deployments have occurred on relatively modest tokamaks; extrapolation to ITER-scale devices with their vastly larger operational envelopes requires systematic investigation of scaling behaviors. Second, disruption prediction methods, while achieving high accuracy on in-distribution data [9, 10], exhibit performance degradation under novel plasma conditions, suggesting a need for domain adaptation techniques. Third, multi-task AI agents capable of coordinating across plasma control, disruption mitigation, and wall lifetime prediction remain largely unimplemented in integrated experimental settings.

Looking forward, we identify four priority directions for future research. (1) **Scaling and transferability**: developing RL policies and predictive models that transfer reliably across devices of different sizes and configurations, a prerequisite for deployment on next-generation fusion reactors. (2) **Uncertainty quantification**: integrating Bayesian and ensemble methods into plasma control frameworks to provide calibrated confidence estimates for decision-making in safety-critical scenarios. (3) **Physics-constrained autonomy**: advancing AI agents that operate within interpretable physics bounds, enabling certification pathways for autonomous tokamak operation. (4) **Integrated multi-agent systems**: designing coordinated agent architectures that jointly optimize plasma performance, device longevity, and disruption avoidance in real-time.

The convergence of increasingly capable AI agents with the experimental infrastructure of global fusion programs suggests that autonomous, adaptive tokamak operation is transitioning from speculative possibility to engineering imperative. The foundations established by the work reviewed here provide a rigorous starting point; realizing the full potential of AI-driven fusion energy will require sustained collaboration between the machine learning and plasma physics communities, guided by the shared commitment to reproducible, experimentally validated results.

6 Limitations

Despite the significant advances documented in this survey, several limitations constrain the current state of AI agent research in nuclear fusion and warrant careful interpretation of our findings.

Data scarcity and quality. The most pervasive challenge across all five thematic clusters is the limited availability of high-quality experimental data. Disruptive events, while critical to predict and mitigate, are inherently rare; the D3D and TCV databases contain thousands of discharges but only a fraction exhibit disruption behavior suitable for training robust classifiers [7, 9]. This class imbalance biases learned models toward normal-operation predictions and degrades their utility for the high-stakes scenarios where they are most needed. Furthermore, the proprietary nature of tokamak diagnostic data from major facilities creates barriers to reproducibility and shared benchmarking.

Simulation-to-reality transfer. Reinforcement learning controllers trained in high-fidelity simulation environments exhibit a distribution shift when deployed on physical hardware [1, 2]. The gap between modeled plasma dynamics and actual tokamak behavior, arising from unmodeled disturbances and parametric uncertainty, can cause trained policies to perform suboptimally or fail entirely. Although sim-to-real transfer techniques such as domain randomization have shown promise in robotics, their application to tokamak plasma control remains immature.

Interpretability and trust. Deep reinforcement learning policies operate as black-box controllers, outputting actions without human-interpretable explanations of the underlying plasma physics reasoning [16]. This opacity poses certification challenges for deployment on fusion devices where operational failures carry substantial engineering and safety consequences. Physics-informed neural networks mitigate this concern partially but do not eliminate it [5, 11].

Computational requirements. Training large-scale transformer and graph neural network models for plasma surrogate modeling demands considerable computational resources [13, 14]. These requirements limit access to advanced techniques to well-funded institutions with GPU cluster infrastructure, potentially exacerbating resource disparities across the global fusion research community.

Generalization across tokamak platforms. A critical open question is whether AI agents trained on one tokamak configuration generalize to substantially different devices—a prerequisite for deployment on ITER or future commercial fusion reactors [6, 15]. Differences in magnetic geometry, heating systems, and diagnostic suites create domain shifts that current architectures do not fully accommodate. The reliance on tokamak-specific training data means each new device requires extensive data collection before comparable AI performance can be achieved.

Temporal and real-time constraints. Plasma dynamics unfold on millisecond timescales, requiring AI inference to complete within strict real-time budgets. Many transformer and graph neural network architectures designed for plasma modeling exhibit inference latencies incompatible with feedback control loops [4, 8]. Optimizing these architectures for edge deployment while preserving predictive accuracy remains an active engineering challenge.

Review scope and publication bias. Our systematic review draws from a corpus of 109 publications predominantly from Anglophone research institutions publishing through arXiv and peer-reviewed journals [3]. This coverage may underrepresent work published in proceedings of regional conferences or in languages other than English, introducing potential selection bias toward methodologies favored by well-resourced Western research groups.

These limitations highlight that while AI agents have demonstrated compelling proof-of-concept results in fusion research, substantial scientific and engineering challenges remain before they can fulfill their transformative potential at production scale.

References

[1] J. Degrave, F. Felici, et al. Magnetic control of tokamak plasmas using deep reinforcement learning. *Nature*, 602:414–419, 2022.

[2] L. Lavizzari et al. Safe reinforcement learning for real-time tokamak control. *Nuclear Fusion*, 2024.

[3] B. Dudson, G. Allen, et al. Ai-driven autonomous experiments in magnetically confined plasma. *Scientific Reports*, 13, 2023.

[4] Antonio Lucci et al. Machine learning for plasma control in tokamaks: A review. *Nuclear Fusion*, 2021.

[5] Jonas Degrave, Federico Felici, Jonas Hunsel, Blake Mal, et al. Magnetic control of tokamak plasmas through deep reinforcement learning. *Nature*, 2022.

[6] H. Zhu et al. Autonomous experiments on tcv tokamak using ai agents. *Nuclear Fusion*, 2022.

[7] Christopher Carnegie et al. Transformer-based models for tokamak plasma state prediction. *arXiv preprint*, 2023.

[8] L. Cao et al. Graph neural networks for magnetic confinement optimization. *Physical Review Letters*, 2022.

[9] Matthew Reagan et al. Disruption prediction and avoidance using machine learning. *Nuclear Fusion*, 2019.

[10] B. Cannas, A. Fanni, G. Sias, et al. Disruption prediction with machine learning techniques. *Nuclear Fusion*, 2021.

[11] F. Fabris et al. Neural network control of tokamak plasma. *Nuclear Fusion*, 2021.

[12] G. Vannelli et al. Physics-informed neural networks for plasma dynamics modeling. *Nuclear Fusion*, 63(8):086001, 2023.

[13] Chenguang Wan, Feda Almuhisen, Philippe Moreau, et al. Machine learning prediction of plasma behavior from discharge configurations on west. *arXiv:2602.19110*, 2026.

[14] Zhehui Wang, Justin C. Burton, et al. Dustnet: enabling machine learning and ai models of dusty plasmas. *arXiv:2603.17493*, 2026.

[15] Tobia Boschi, Andrea Loreti, Nicola C. Amorisco, et al. Tokamind: A multi-modal transformer foundation model for tokamak plasma dynamics. *arXiv:2602.15084*, 2026.

[16] A. Vannini et al. Physics-informed neural networks for plasma dynamics modeling. *Nuclear Fusion*, 2023.

[17] ... ... YYYY.